

Research Interests

Embedded systems; Wearable and Ubiquitous systems; Internet of Things (IoT); Embedded Cybersecurity; Machine Learning; Assistive Technology Design; Reconfigurable Computing

Education

- **University of Arkansas** Fayetteville, AR
B.S. in Electrical Engineering August 2022-Present
 - GPA: 3.93/4.0
 - Minor in Mathematics
- **University of Oxford, Oriel College** Oxford, UK
Study Abroad Summer 2023
 - Completed 3 US Credits/6 ECTS course on Quantum Computing that included 45 total hours of workshops, lectures, projects, and seminars.

Research Experience

- **Inan Research Lab** Atlanta, GA
SURE Program, Georgia Institute of Technology May 2025-present
 - Principal Investigator: Omer T. Inan
 - Working to develop deep learning models for prediction of blood volume decomposition status (BVDS) based on ECG, SCG, and PPG signals from wearable sensing device on pig hearts.
 - Reading scientific literature, understanding the context of a research question, using a deep learning framework to code, train, validate, and test an AI model with minor supervision, and will interpret and present my results in an oral and written form.
 - Attending seminars and will present poster at program final research symposium.
- **Applied Embedded Systems and IoT Research Lab (ÆSIR)** Fayetteville, AR
Undergraduate Research Assistant, University of Arkansas August 2023-present
 - Principal Investigator: Alexander H. Nelson
 - Conducted research in embedded systems and hardware security. Focus areas include capacitive sensor arrays, fault injection on SRAM PUFs, and secure microcontroller architectures.
 - Designed and tested custom PCBs, implemented embedded C firmware across MSP430 and ESP32 platforms, and utilized FPGA-based acceleration for real-time data processing.
 - Contributed to peer-reviewed conference publications; awarded funding through the Honors College Research Grant and Goldwater Scholarship.
- **GRAPES Research Group** Fayetteville, AR
Honors Research Experience, University of Arkansas August 2022-May 2023
 - Principal Investigator: Roy A. McCann
 - Developed a wireless circuit utilizing an ATmega328 microprocessor to measure Electric and Magnet Fields and Temperatures of power lines, enabling cost-effective monitoring for local power grids and relaying data to a centralized database. Awarded Best Poster prize at the 2023 Honors Research Symposium

Peer-reviewed Publications

- P2:** A. Komanduri and A. H. Nelson, "Power PUFs: Strengthening SRAM PUFs Against Fault Injection on Low-Cost IoT Devices," in 2025 IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events (PerCom Workshops), Washington DC, DC, USA, 2025, pp. 665-668, doi: 10.1109/PerComWorkshops65533.2025.00158.
- P1:** A. Komanduri and A. Nelson, "Daisy-Chaining Embedded Processors for Enhanced Capacitive Sensor Array Resolution," 2024 IEEE Green Technologies Conference (GreenTech), Springdale, AR, USA, 2024, pp. 166-167, doi: 10.1109/GreenTech58819.2024.10520578.

Presentations and Posters

- T5:** “Predicting Hemodynamic Blood Volume Loss based on Multi-Channel Physiological Signals using Deep Learning“, 2025 Georgia Tech SURE Research Symposium, Atlanta, GA, July 2025. (presentation + poster)
- T4:** “Power PUFs: Strengthening SRAM PUFs Against Fault Injection on Low-Cost IoT Devices“, 2025 IEEE International Conference on Pervasive Computing and Communications (PerCom), Washington, D.C. March 2025. (poster)
- T3:** “Daisy-Chaining Embedded Processors for Enhanced Capacitive Sensor Array Resolution“, 2024 IEEE Green Technologies Conference (GreenTech), Springdale, AR. April 2024. (poster)
- T2:** “An Analysis of a Finite Element Based Framework for Capacitive Sensing Simulations“, 85th Annual Meeting of OK-AR MAA Section, Fayetteville, AR. April 2024. (presentation)
- T1:** “Designing Embedded Systems to Monitor EMFs and Power Grids in Real-Time“, 15th Annual Honors Engineering Symposium, Fayetteville, AR. April 2022. (presentation + poster)

Selected Honors and Awards

Barry Goldwater Scholarship	Mar. 2025
Honors College Travel Grant	Feb. 2025
Honors College Research Grants	Jan. 2024/25
Best Poster Award, Honors Research Symposium	Apr. 2023
Honors College Fellowship	Aug. 2022
Arkansas Governor’s Distinguished Scholarship	Aug. 2022

Professional Experience

- **Wolfspeed Inc.** Fayetteville, AR
Power Modules Backend Operations Intern May 2024-August 2024
 - Worked with Power Modules Test Engineering team to design high reliability modules for electrical testing on Wolfspeed Power Modules to run a Gage R&R study.
 - Analyzed past module test data using JMP to compile module failure trends.
 - Coordinated with offshore assembly and test sites to streamline the flow of incoming test data and created dashboards using PowerBI.
- **University of Arkansas, Mathematical Sciences** Fayetteville, AR
Grading Assistant August 2023-May 2024
 - Graded homeworks, quizzes, and exams for MATH 1203 (College Algebra) and MATH 2584 (Elementary Differential Equations). Interacted with course instructors to create rubrics and a fair grading scale.

Selected Leadership and Service

- **UARK IEEE Chapter**
President January 2024 – Present
 - Organize student events for Electrical Engineering and Computer Science students to promote inclusion and give students networking opportunities
 - Founder and director of UARK’s first student-led, student-organized research conference made for the students, by the students
- **REC Foundation/VIQC Robotics**
Head Referee October 2022-Present
 - Refereed at local middle school robotics competitions and managed hundreds of matches, assigned scores, and made difficult decisions.